

The strict decay condition in Proposition 3.3 is not necessary

A critical-endpoint theorem for the scalar Dirac operator

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Abstract

Stahlhut asked whether the strict condition $\tau > n/p - n/q$ in his L^p – L^q off-diagonal functional-calculus theorem is necessary. We give a negative answer in general. For the one-dimensional unperturbed scalar Dirac operator $D = -i d/dx$, the entire conclusion holds at the critical exponent $\tau = 1/p - 1/q$, uniformly for every admissible symbol ψ and every bounded holomorphic multiplier g . In fact the resulting off-diagonal order is stronger than the order asserted in the source proposition. An explicit symbol with exact critical decay shows that this is not an artifact of nested symbol classes.

1. Exact source question

The source defines $\Psi_\sigma^\tau(\dot{S}_\nu)$ by the estimate

$$|\psi(z)| \lesssim \frac{|z|^\sigma}{1 + |z|^{\sigma+\tau}} \quad (z \in \dot{S}_\nu),$$

where $\dot{S}_\nu$ is the union of the two open sectors about the positive and negative real axes. Proposition 3.3 states that, for $p < q$ in the admissible interval and $\tau > n/p - n/q$, the families $\{g(T)\psi(tT)\}_{t>0}$ satisfy L^p – L^q off-diagonal estimates of all orders $0 \leq K < \sigma/c_{p,q}$. The immediately following remark reads:

“We do not know if the condition $\tau > \frac{n}{p} - \frac{n}{q}$ is necessary in Proposition 3.3.”

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Remark 3.8. *We do not know if the condition $\tau > \frac{n}{p} - \frac{n}{q}$ is necessary in Proposition 3.3.*

Sometimes, it is appropriate to have the following variant of Proposition 3.3. See for example [Aus-Sta-2].

Proposition 3.9 (L^p – L^q biparameter off-diagonal estimates). *Let $T \in \{DB, BD\}$.*

For clarity, a family $\{A_t\}_{t>0}$ has L^p – L^q off-diagonal estimates of order K when

$$\|\mathbf{1}_E A_t(\mathbf{1}_F f)\|_q \leq C t^{1/q-1/p} \left(1 + \frac{\text{dist}(E, F)}{t}\right)^{-K} \|\mathbf{1}_F f\|_p$$

for all Borel sets $E, F \subset \mathbb{R}$, all $t > 0$, and all $f \in L^p$.

2. Result and proof intuition

Fix $1 < p < q < \infty$ and put

$$\delta = \frac{1}{p} - \frac{1}{q} \in (0, 1).$$

At high frequency, a critical symbol has order $-\delta$. Its Fourier kernel therefore has the Riesz-potential singularity $|x|^{\delta-1}$, which is exactly the Hardy–Littlewood–Sobolev kernel mapping L^p to L^q . At low frequency, the zero of order σ gives the much faster far-field decay $|x|^{-1-\sigma}$. The first behavior gives endpoint boundedness; the second gives off-diagonal decay. Holomorphy supplies all annular derivative bounds uniformly, even after inserting the unscaled factor $g(D)$.

Theorem 1 (Uniform critical endpoint). *Let $D = -i d/dx$ on scalar functions on $\mathbb{R}$, let $B = I$, and fix $0 < \nu < \pi/2$, $\sigma > 0$, and $1 < p < q < \infty$. Set $\delta = 1/p - 1/q$. For every*

$$\psi \in \Psi_\sigma^\delta(\dot{S}_\nu) \quad \text{and} \quad g \in H^\infty(\dot{S}_\nu),$$

the family $\{g(D)\psi(tD)\}_{t>0}$ satisfies L^p – L^q off-diagonal estimates of every order

$$0 \leq K \leq \sigma + \delta.$$

The constants are uniform in t , depend on ψ through its defining Ψ_σ^δ bound, and depend on g only through $\|g\|_\infty$.

Proof. We use the Fourier transform convention for which D has symbol ξ . The multiplier of $g(D)\psi(tD)$ is

$$g(\xi)\psi(t\xi) = h_t(t\xi), \quad h_t(\eta) := g(\eta/t)\psi(\eta).$$

We first prove a kernel estimate uniform in t . On either real half-axis, a disk of radius $c_\nu|\eta|$ about η remains inside the corresponding component of $\dot{S}_\nu$. Cauchy's inequalities, the boundedness of g , and the defining estimate for ψ imply, for every integer $m \geq 0$,

$$|h_t^{(m)}(\eta)| \leq C_m \|g\|_\infty |\eta|^{-m} \begin{cases} |\eta|^\sigma, & 0 < |\eta| \leq 1, \\ |\eta|^{-\delta}, & |\eta| \geq 1. \end{cases} \quad (1)$$

The constants do not depend on t : positive dilation preserves each sector, so $z \mapsto g(z/t)$ has the same supremum bound on it.

Choose $\varphi \in C_c^\infty(\{1/2 \leq |\eta| \leq 2\})$ such that $\sum_{j \in \mathbb{Z}} \varphi(2^{-j}\eta) = 1$ for $\eta \neq 0$, and put

$$h_{t,j}(\eta) = \varphi(2^{-j}\eta)h_t(\eta), \quad \kappa_{t,j} = \mathcal{F}^{-1}h_{t,j}.$$

Scaling and M integrations by parts, using (1), give

$$|\kappa_{t,j}(x)| \leq C_M 2^j a_j (1 + 2^j |x|)^{-M}, \quad a_j = \begin{cases} 2^{j\sigma}, & j \leq 0, \\ 2^{-j\delta}, & j > 0. \end{cases} \quad (2)$$

This standard estimate can also be read directly after rescaling the annulus to unit size: all derivatives of the rescaled multiplier are $O(a_j)$, and Fourier integration by parts supplies the last factor.

The series of L^1 norms is bounded by $C \sum_j a_j < \infty$. Hence $\kappa_t := \sum_j \kappa_{t,j}$ is an L^1 kernel representing h_t . Summing (2), and splitting at $2^j |x| \simeq 1$, yields

$$|\kappa_t(x)| \leq C \begin{cases} |x|^{\delta-1}, & 0 < |x| \leq 1, \\ |x|^{-1-\sigma}, & |x| \geq 1. \end{cases} \quad (3)$$

For completeness, when $|x| \leq 1$, the low-frequency sum is bounded and the high-frequency sum is $O(|x|^{\delta-1})$. When $|x| \geq 1$, the low-frequency sum is $O(|x|^{-1-\sigma})$; choosing $M > 1 + \sigma$ makes the high-frequency sum smaller. All bounds remain uniform in t .

The kernel of $g(D)\psi(tD)$ is therefore

$$K_t(x) = t^{-1}\kappa_t(x/t).$$

Since (3) implies $|K_t(x)| \leq Ct^{-\delta}|x|^{\delta-1}$, the one-dimensional Hardy–Littlewood–Sobolev inequality gives

$$\|K_t * f\|_q \leq Ct^{-\delta}\|f\|_p. \quad (4)$$

It remains to retain decay between separated sets. Let $\rho = \text{dist}(E, F)$ and let r satisfy

$$1 + \frac{1}{q} = \frac{1}{p} + \frac{1}{r}, \quad \text{so that} \quad \frac{1}{r} = 1 - \delta.$$

If $\rho \geq t$, (3) and scaling give

$$\begin{aligned} \|\mathbf{1}_{\{|x| \geq \rho\}} K_t\|_r &= t^{1/r-1} \|\mathbf{1}_{\{|u| \geq \rho/t\}} \kappa_t\|_r \\ &\leq Ct^{-\delta}(\rho/t)^{-\sigma-\delta}. \end{aligned}$$

Young's convolution inequality applies because $x \in E$, $y \in F$ implies $|x - y| \geq \rho$. It thus supplies the desired estimate for every $K \leq \sigma + \delta$ when $\rho \geq t$. When $\rho < t$, the global bound (4) supplies it, since $(1 + \rho/t)^{-K}$ is bounded below by 2^{-K} . This proves the theorem. $\square$

3. Genuine critical decay and comparison with Proposition 3.3

The critical class is genuinely reached. On each component of $\dot{S}_\nu$ define

$$w(z) = \text{sgn}(\Re z)z, \quad \psi_0(z) = \frac{w(z)^\sigma}{(1 + w(z))^{\sigma+\delta}},$$

using the principal powers. Since w lies in the right sector, $|1 + w| \simeq 1 + |w|$, so $\psi_0 \in \Psi_\sigma^\delta(\dot{S}_\nu)$. Along the positive axis,

$$\lim_{s \rightarrow \infty} s^\delta \psi_0(s) = 1.$$

Consequently ψ_0 belongs to no $\Psi_\sigma^{\tau'}(\dot{S}_\nu)$ with $\tau' > \delta$.

For $B = I$, the source records $\mathcal{I}_{D,I} = (1, \infty)$. Hence in dimension one its printed constant becomes

$$c_{p,q} = (1 - (1/p - 1/q))^{-1} = (1 - \delta)^{-1},$$

and Proposition 3.3 asks for every $K < \sigma/c_{p,q} = \sigma(1 - \delta)$. Our endpoint theorem gives every $K \leq \sigma + \delta$, and $\sigma(1 - \delta) < \sigma + \delta$. Thus the full conclusion of Proposition 3.3 holds in this admissible model at equality $\tau = n/p - n/q$, even for a symbol having no extra decay.

This supplies a full negative answer to the necessity question as stated. It does *not* prove that the endpoint holds for every rough perturbation DB or BD ; that stronger universal endpoint problem remains open.

4. Verification and novelty boundary

The proof uses only: (i) Fourier diagonalization of $-i d/dx$; (ii) Cauchy estimates on sectors; (iii) a smooth dyadic partition and Fourier integration by parts; (iv) the one-dimensional Hardy–Littlewood–Sobolev inequality; and (v) Young’s convolution inequality. No numerical evidence is used. The two exponents in (3) have independent roles: $\delta - 1$ gives the exact L^p – L^q endpoint, while $-1 - \sigma$ gives the tail power $\sigma + \delta$.

The run registry and bounded primary-source searches through August 11, 2026 found no later paper claiming this resolution. Priority is not asserted; specialist citation review is recommended. Human review should focus on the uniformity of (1) under $g(\eta/t)$, the dyadic summation in (3), and the intended logical reading of “necessary” in the source remark.

References

- [1] S. Stahlhut, *L^p – L^q theory for holomorphic functions of perturbed first order Dirac operators*, arXiv:1403.5368 (2014), Proposition 3.3 and Remark 3.8.
- [2] E. M. Stein, *Singular Integrals and Differentiability Properties of Functions*, Princeton University Press, 1970; see the Hardy–Littlewood–Sobolev theorem.